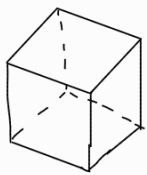
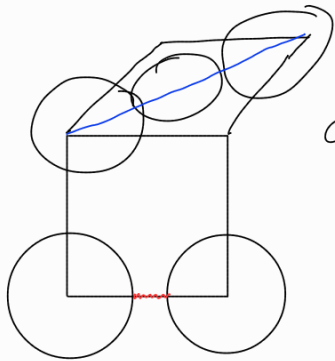
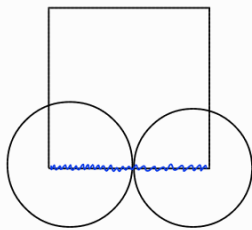


$Ar \Rightarrow \frac{1}{4}$
 $S \Rightarrow \frac{1}{8}$
 $F \Rightarrow \frac{1}{2}$
 $C \Rightarrow 1$



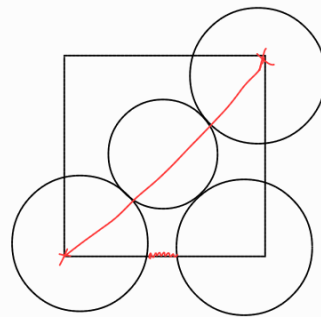
$S \Rightarrow 8$
 $F \Rightarrow 6$
 $Ar \Rightarrow 12$
 $C \Rightarrow 1$



$$a\sqrt{2} = 2r_1 + 2r_2$$

$$a = \frac{2r_1 + 2r_2}{\sqrt{2}}$$

CFC



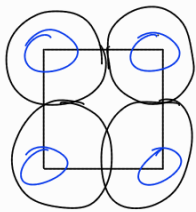
$$a = \frac{2r_1 + 2r_2}{\sqrt{3}}$$

$$a\sqrt{3} = 2r_1 + 2r_2$$

CC

$$\rho = \frac{m}{V}$$

$\swarrow$ M



C

$$S = \frac{1}{8} \times 8 = 1$$

1 atome

CC

$$S = \frac{1}{8} \times 8 = 1$$

$$C \Rightarrow 1 \times \frac{1}{2} = 1$$

2 atomes

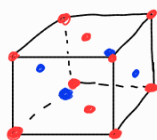
CFC

$$S \Rightarrow \frac{1}{8} \times 8 = 1$$

$$F \Rightarrow \frac{1}{2} \times 6 = 3$$

4 atome

Ex #9



$$\begin{aligned} S &= \frac{1}{8} \times 8 = 1 \\ f &= \frac{1}{2} \times 2 = 1 \\ C &= 1 \times 0 = 0 \\ Ar &= \frac{1}{4} \times 0 = 0 \end{aligned}$$

$$\underline{Fe = 2}$$

$$\begin{aligned} S &= \frac{1}{8} \times 0 = 0 \\ f &= \frac{1}{2} \times 4 = 2 \\ C &= 1 \times 0 = 0 \\ Ar &= \frac{1}{4} \times 0 = 0 \end{aligned}$$

$$\underline{Pd = 2}$$

$$a) X_n = \frac{n_{Pd}}{n_{Pd} + n_{Fe}} = \frac{2}{4} = \frac{1}{2}$$

$$b) \rho = \frac{m}{V}$$

$$m = \left(n_{Pd} \times \frac{M_{Pd}}{N_A} \right) + \left(n_{Fe} \times \frac{M_{Fe}}{N_A} \right)$$

$$m = \left(2 \times \frac{106.4}{6.022 \cdot 10^{23}} \right) + \left(2 \times \frac{55.85}{6.022 \cdot 10^{23}} \right) = 5.38868 \cdot 10^{-22}$$

$$V \Rightarrow a^3 \Rightarrow a\sqrt{2} = 4 \cdot 175 \text{ pm} \times 10^{-12} \text{ m}$$

$$a = \left(\frac{4 \cdot 175 \cdot 10^{-12}}{\sqrt{2}} \right)^3$$

Ex #11

$$\begin{array}{l} C \\ S \frac{1}{8} \times 8 = 1 \\ f \frac{1}{2} \times 6 = 3 \\ C 1 \times 0 = 0 \\ Ar \frac{1}{4} \times 0 = 0 \end{array}$$

$$\begin{array}{l} Zr \\ \times 0 = 0 \\ \times 0 = 0 \\ \times 1 = 1 \\ \times 12 = 3 \end{array}$$

$$\underline{4}$$

$$\underline{4}$$

$$\rho = \frac{m}{V} = \frac{\frac{4}{6.022 \cdot 10^{23}} \times (91.22 + 12.01)}{(4r)^3} = 6.73 \text{ g/cm}^3$$

$$\Rightarrow a = 4.67 \cdot 10^{-8} \text{ cm}$$



$$b) r = 1.17 \cdot 10^{-8} \text{ cm}$$

$$c) 4 \times \frac{2}{100} = 0.08$$

Exercice Crystallins

$$1) A \text{ rayon} = 25.4 \text{ mm} / 2 = 12.7 \text{ mm}$$

$$C \text{ rayon} = 50.6 \text{ mm} / 2 = 25.3 \text{ mm}$$

$$2 \cdot 12.7 + 2 \cdot 25.3 = 76 \text{ mm}$$

2)

Fe	O
$\text{C } 1 \times 1 = 1$	$\times 0 = 0$
$\text{f } \frac{1}{2} \times 0 = 0$	$\times 6 = 3$
$\text{S } \frac{1}{8} \times 0 = 0$	$\times 8 = 1$
$\text{Ar } \frac{1}{4} \times 12 = 3$	$\times 0 = 0$
4	4

4 de chaque ion dans une maille

3) $140 + 75 + 75 + 140 = 430 \cdot 10^3 \text{ nm}$

4)
$$\frac{f_c = 4 \times \left(\frac{4}{3} \cdot \pi \cdot R_{\text{Na}^+}^3 \right) + 4 \times \left(\frac{4}{3} \cdot \pi \cdot R_{\text{Cl}^-}^3 \right)}{a^3}$$

$$\frac{f_c = 4 \times \left(\frac{4}{3} \cdot \pi \cdot 75^3_{\text{Fe}} \right) + 4 \times \left(\frac{4}{3} \cdot \pi \cdot 140^3_{\text{O}} \right)}{(430)^3} = 0,67$$

5)

$$\frac{4 \cdot (55,85 + 16)}{6,022 \cdot 10^{23}}$$

$$\left(430_{\text{pm}} \cdot 10^{-10}_{\text{cm}} \right)^3$$